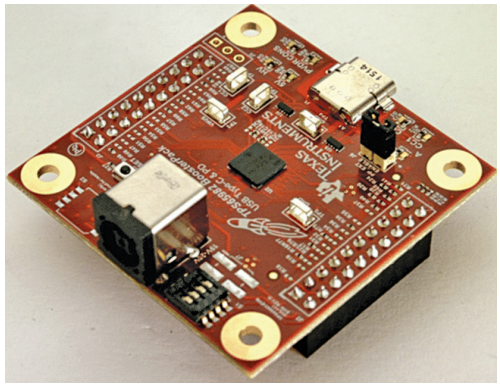
Applications. The reference design is intended for applications such as smartphones, tablets, and other mobile devices requiring fast charging and bidirectional USB Type-C functionality.

Link. *https://efy.co.in/kkhj2n*

OEM Brand. Monolithic Power Systems (MPS)

USB Type-C Power Delivery

The reference design from Texas Instruments can become a starting



point for you to develop USB Type-C and Power Delivery products that need power management, data connectivity, and alternate-mode support such as DisplayPort. It includes a USB power delivery controller that supports USB Type-C compliance and can switch between source and sink modes, allowing a device to either supply power or receive power. The design also includes over-voltage and over-current protection, a data-port multiplexer for routing high-speed USB and alternate-mode signals, and a USB low-speed endpoint. A DC/DC converter provides regulated power, while ESD protection and power switching components support the interface and power-management functions.

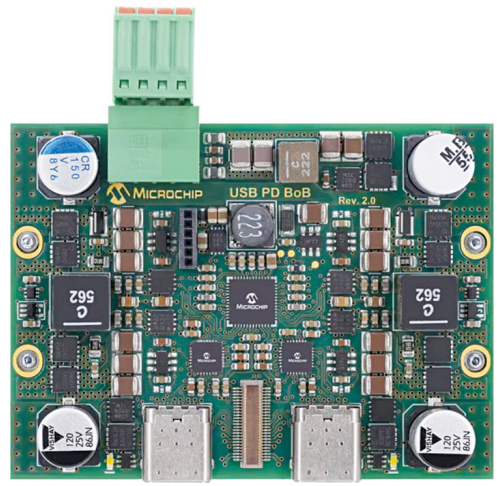
Applications. The reference design is suitable for products such as all-in-one PC motherboards, monitors, gaming PCs, and standard PC motherboards.

Link. *https://efy.co.in/cvl1k3*

OEM Brand. Texas Instruments (TI)

USB-C Power Delivery

Microchip Technology's USB-C Power Delivery (USB-PD) reference design helps build charging systems for multiple USB-C devices. It supports two USB-C ports, delivering up to 60W per port and 90W when both are used, with power distributed based on the connected devices. It supports common USB-PD voltage levels and PPS, along with overvoltage, overcurrent, short-circuit, and temperature protection. A GUI allows designers to monitor input, output, and temperature, and adjust settings during testing. The power-conversion section can be modified for different system requirements, while the design can be expanded to support up to four USB-PD ports. It also includes con-



troller area network (CAN) and local interconnect network (LIN) interfaces for automotive systems, offers up to 97 per cent efficiency, and provides an option for electromagnetic interference reduction.

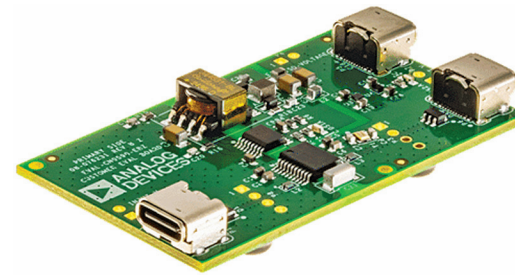
Applications. For automotive applications, the design can integrate with vehicle systems through CAN and LIN interfaces.

Link. *https://efy.co.in/95wmsm*

OEM Brand. Microchip Technology

USB Type-C connector

The reference design from Analog Devices helps you safely connect USB 2.0 devices while keeping the host and device electrically separate. It supports data speeds of up to 480Mbps and can provide up to 2.5W of isolated power. The design can get power directly from the USB



Type-C cable, so no separate power supply is needed. A second USB Type-C port can provide extra power when the connected device needs more than 2.5W. It also protects against reverse polarity, overvoltage, and overcurrent. Galvanic isolation helps break ground loops, reduce the effects of electrical disturbances, improve noise immunity, and prevent signal problems.

Applications. You can use this design in industrial, energy, measurement, and medical systems to reduce noise, ground loops, and high-voltage risks.

Link. *https://efy.co.in/up459w*

OEM Brand. Analog Devices (ADI)

If you are working on a USB-C power or charging product, you can use these reference designs as a starting point instead of designing everything from scratch. Choose a design that is close to your needs and modify it for your product. This can help you test your ideas faster and save development time. EFY

Nidhi Agarwal is Senior Technology Journalist at EFY with a deep interest in embedded systems, development boards, and IoT cloud solutions.